

Customer Churn

961731 Business Data Analytics

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Agenda

1. Business

What is churn worth?

2. EDA

What does the data say?

3. Features

How does behavior change?

4. Models

Who is at risk?

5. Advanced ML

How do we improve robustness?

6. Causal action

What should we do?

Module 1: The Business of Churn and Data Intuition

MODULE 1

Define the loss, the clock, the value, and the action before touching an algorithm.

Define churn

Measure loss

Value customers

Specify action

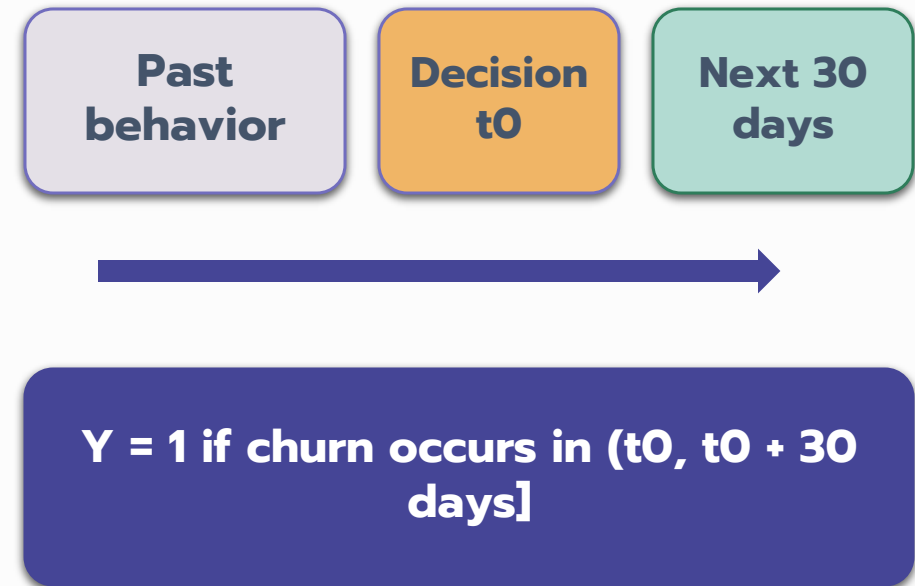
Verizon framed retention as an operational decision

- **Business problem:** customers may leave after a poor service interaction.
- **Data-to-action bridge:** predict call intent and match the customer to a suitable agent.
- **Important caveat:** reported targets are not the same as measured causal outcomes.



Churn is a label defined by a clock and a rule

- **Contract churn:** service is canceled or not renewed.
- **Behavioral churn:** no meaningful activity for a defined number of days.
- **Prediction target:** Will this active customer churn within the next 30 days?



Three conventional measures answer different questions

Logo churn

**Lost customers ÷ customers
at start**

How many relationships
ended?

Revenue churn

**Lost recurring revenue ÷
starting revenue**

How much recurring money
disappeared?

Cohort retention

**Active original cohort ÷
starting cohort**

How well does one intake
survive?

Classification: count-based · money-weighted · cohort-based

A growing customer base can still hide 12 percent churn

Start
100

Lost
12

Acquired
20

End
108

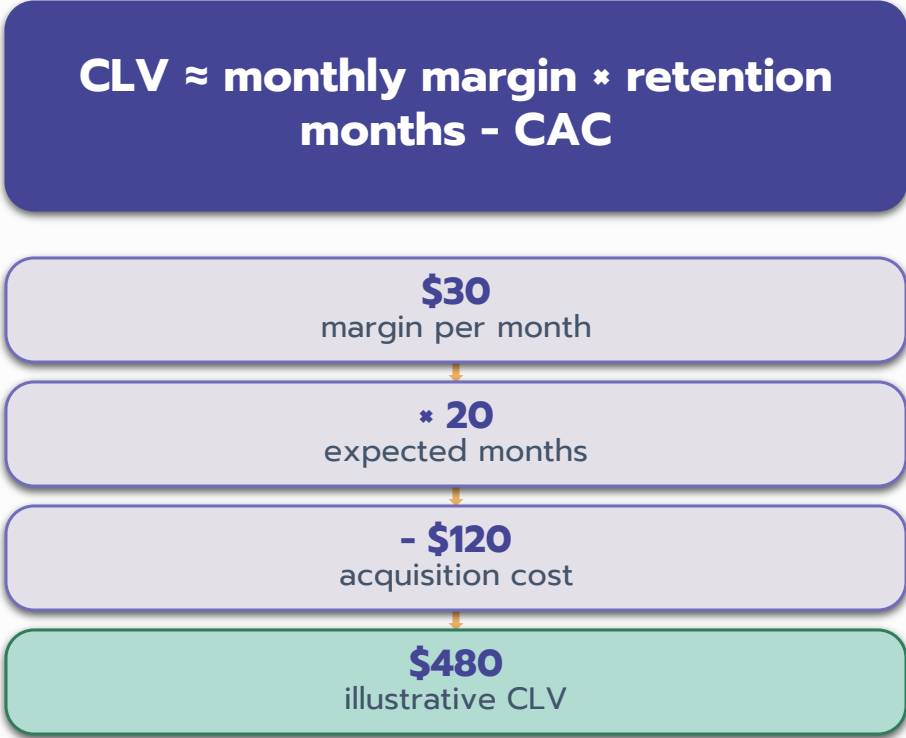
Logo churn = $12 \div 100 = 12\%$

Net growth = $(108 - 100) \div 100 = 8\%$

Growth and churn can be true at the same time.

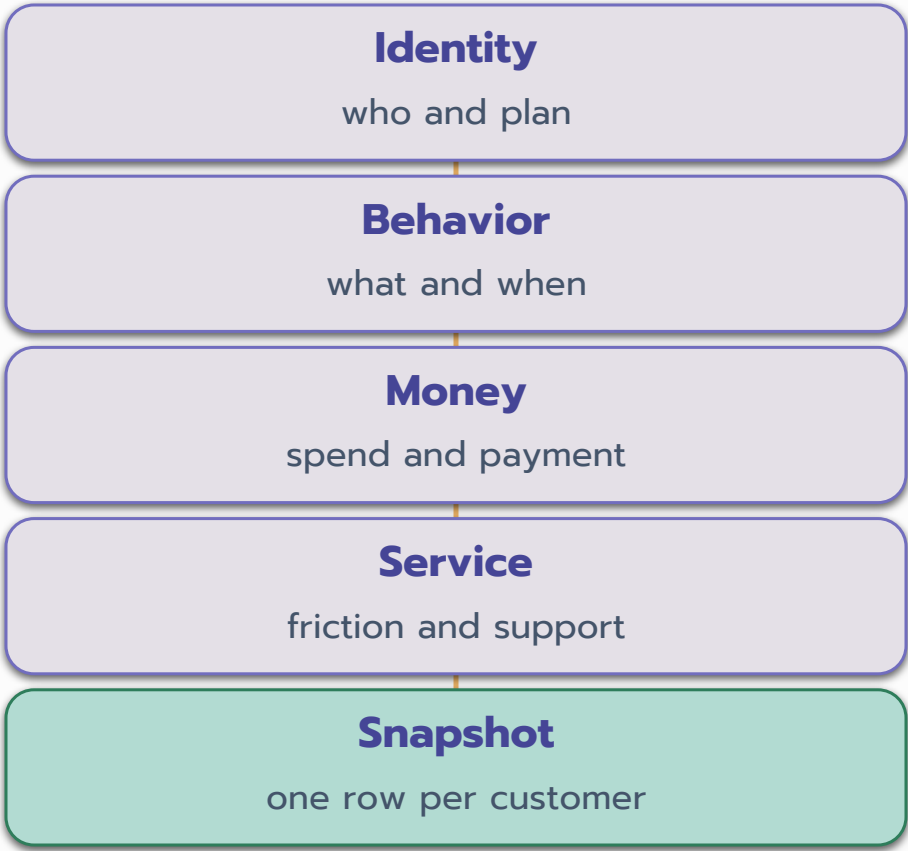
CLV converts retention into a money decision

- **Customer lifetime value:** expected future contribution after acquisition cost.
- **Decision rule:** spend on retention only when the expected saved value exceeds the intervention cost.
- **Illustrative customer:** \$30 monthly margin, 20 expected months, \$120 acquisition cost.



Useful churn data spans identity, behavior, money, and service

- **Identity and plan:** tenure, contract, geography, product mix.
- **Behavior and transactions:** sessions, usage, purchases, payments.
- **Service and sentiment:** tickets, wait time, complaints, survey text.
- **Operational metrics:** coverage, refresh latency, missingness, and action capacity.



Module 2: Exploratory Data Analysis and Data Wrangling

MODULE 2

Turn raw rows into a trustworthy story before asking a model to learn.

Inspect rows

Check quality

Visualize patterns

Block leakage

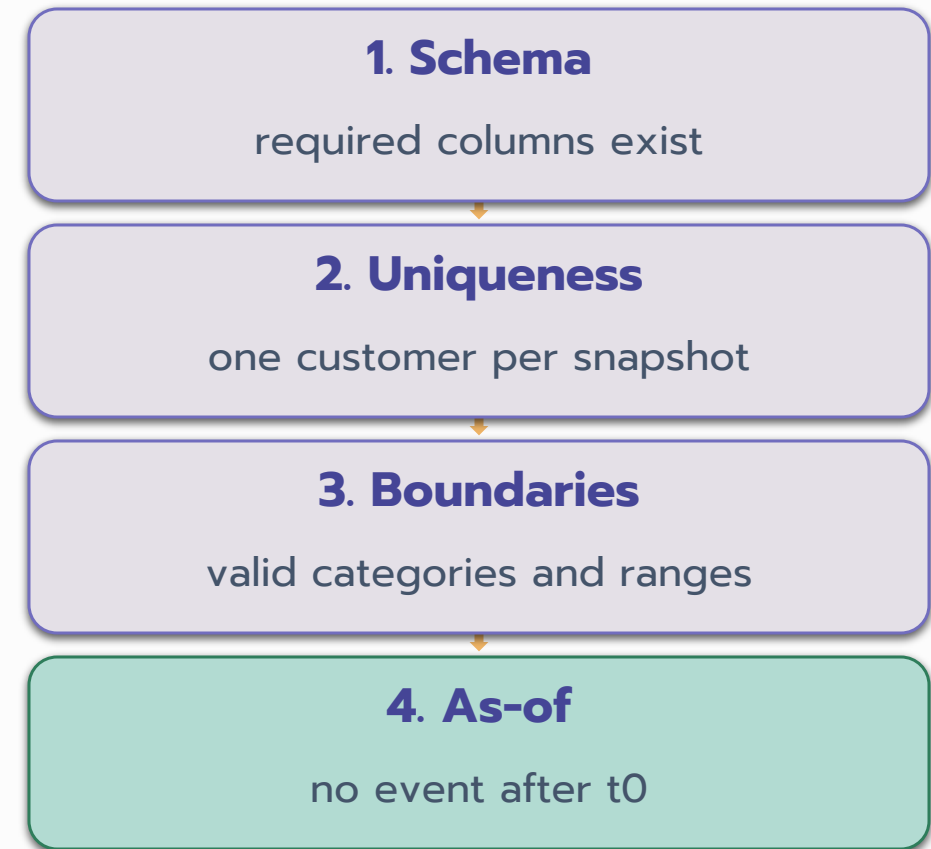
A clean eight-customer table will anchor every method

ID	Tenure	Recency	Logins	Tickets	Contract	Monthly	Churn
C01	2	14	4	3	Monthly	\$85	Yes
C02	18	1	22	0	1-year	\$65	No
C03	6	8	8	2	Monthly	\$92	Yes
C04	36	0	30	0	2-year	\$55	No
C05	12	3	16	1	Monthly	\$75	No
C06	1	21	1	4	Monthly	\$98	Yes
C07	24	2	20	0	1-year	\$70	No
C08	9	10	6	2	Monthly	\$88	Yes

Teaching sample: Recency = days since last activity; Logins and Tickets use the last 30 days.

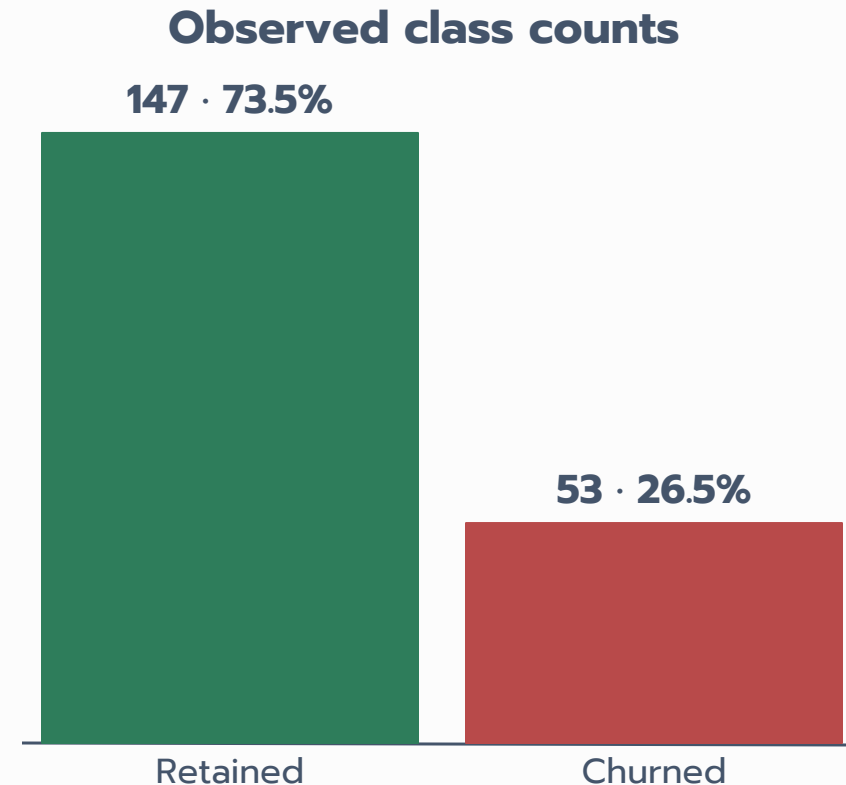
Before modeling, verify meaning, type, range, and time

- **Meaning:** one row means one customer at one snapshot.
- **Type:** dates are dates, numbers are numeric, categories are explicit.
- **Range:** tenure and recency cannot be negative.
- **Time:** every feature timestamp must be earlier than t0.



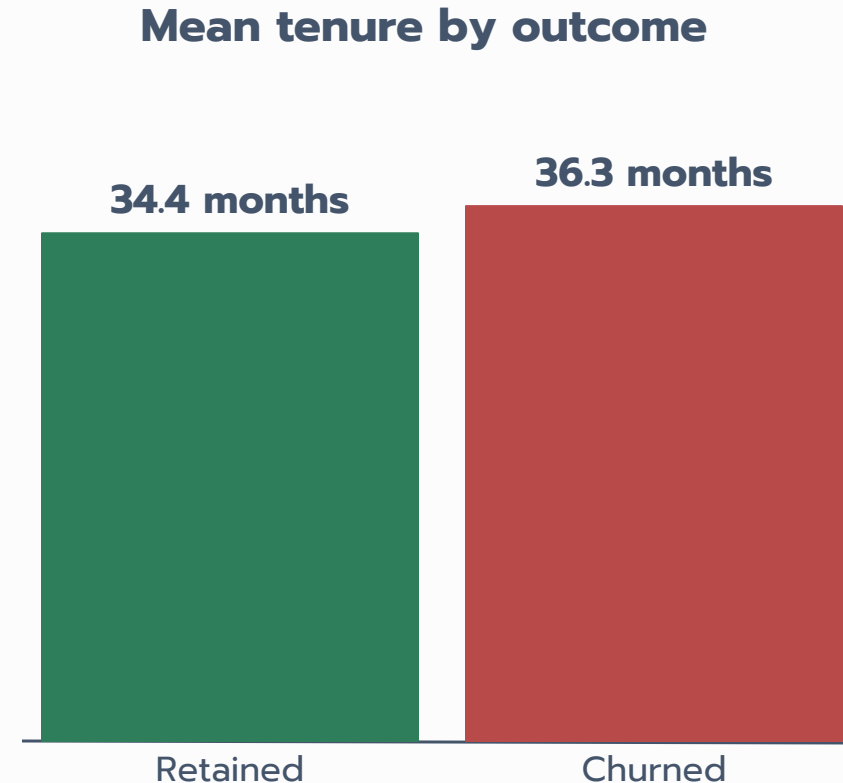
Class balance sets the baseline a model must beat

- **Retained:** 147 of 200 customers (73.5%).
- **Churned:** 53 of 200 customers (26.5%).
- **Naive baseline:** always predict stay and obtain 73.5% accuracy with 0% churn recall.



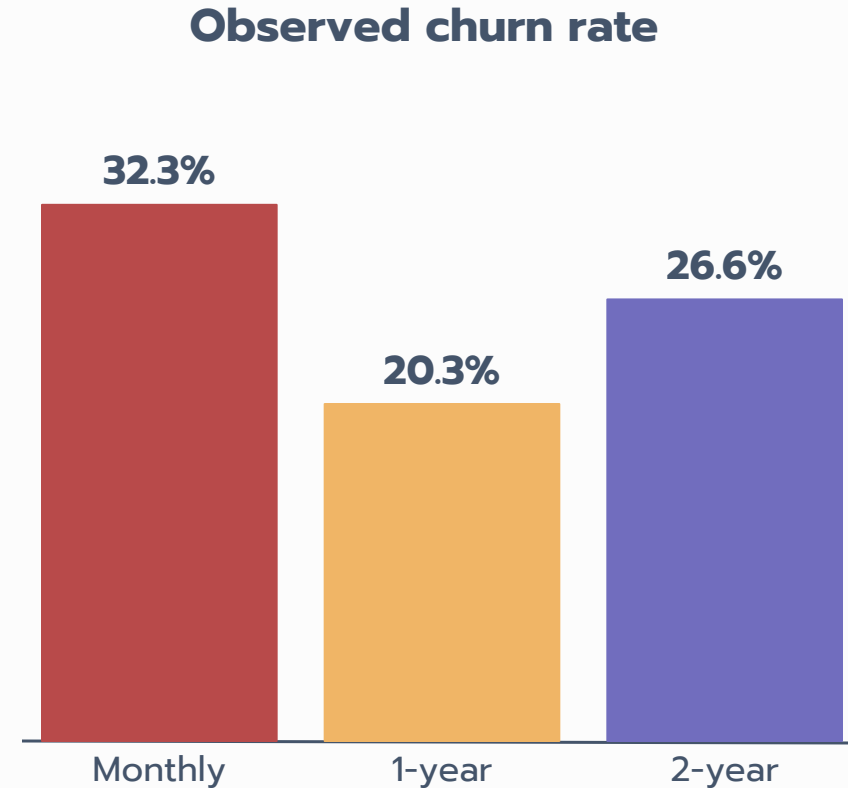
Tenure alone does not separate churners in this sample

- **Retained mean tenure:** 34.4 months.
- **Churned mean tenure:** 36.3 months.
- **Interpretation:** heavy overlap means tenure alone is a poor separator here.
- **ML effect:** combine tenure with contract, payment, service, and behavior.



Contract type carries a stronger signal than a single charge

- **Month-to-month:** 32.3% observed churn.
- **One-year:** 20.3% observed churn.
- **Two-year:** 26.6% observed churn.
- **ML effect:** encode contract and test interactions rather than forcing one threshold.



A chi-square table is easy to compute but unsafe here

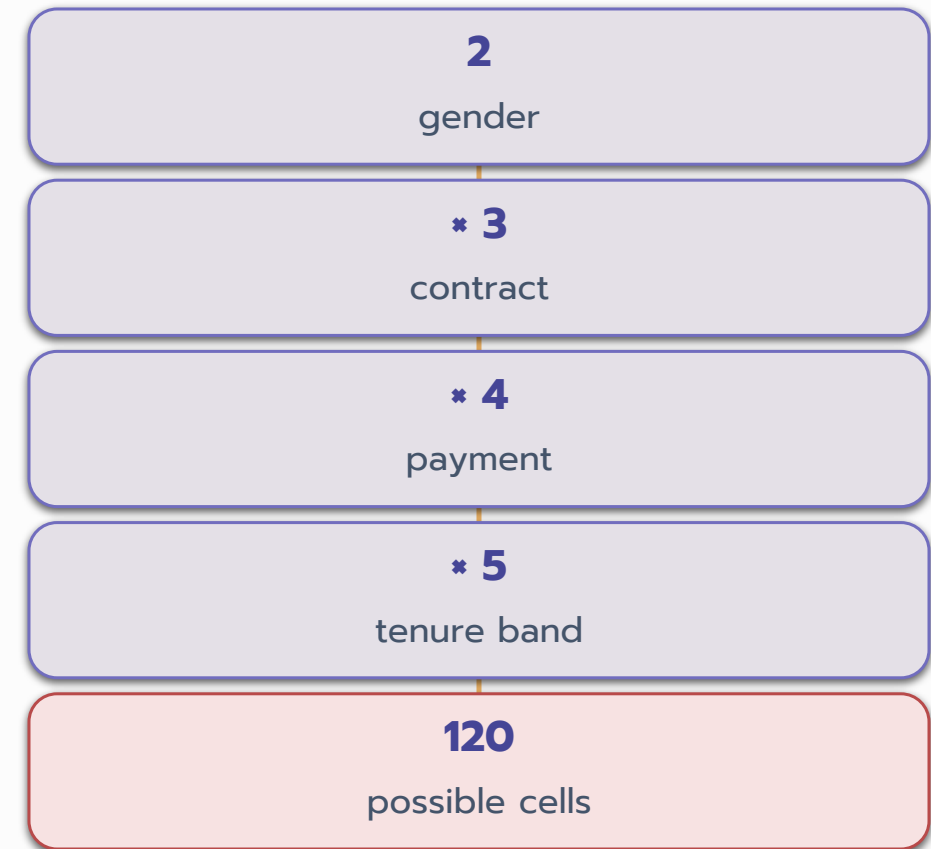
- **Observed table:** monthly 4 churn / 1 stay; longer contracts 0 churn / 3 stay.
- **Expected counts:** range from 0.5 to 2.5 under independence.
- **Computed statistic:** $\chi^2 = 4.8$ with 2 degrees of freedom.
- **Stop condition:** the large-sample approximation is unreliable because expected counts are below 5.

Contract	Churn	Stay
Monthly	4	1
1-year	0	2
2-year	0	1

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}} = 4.8$$

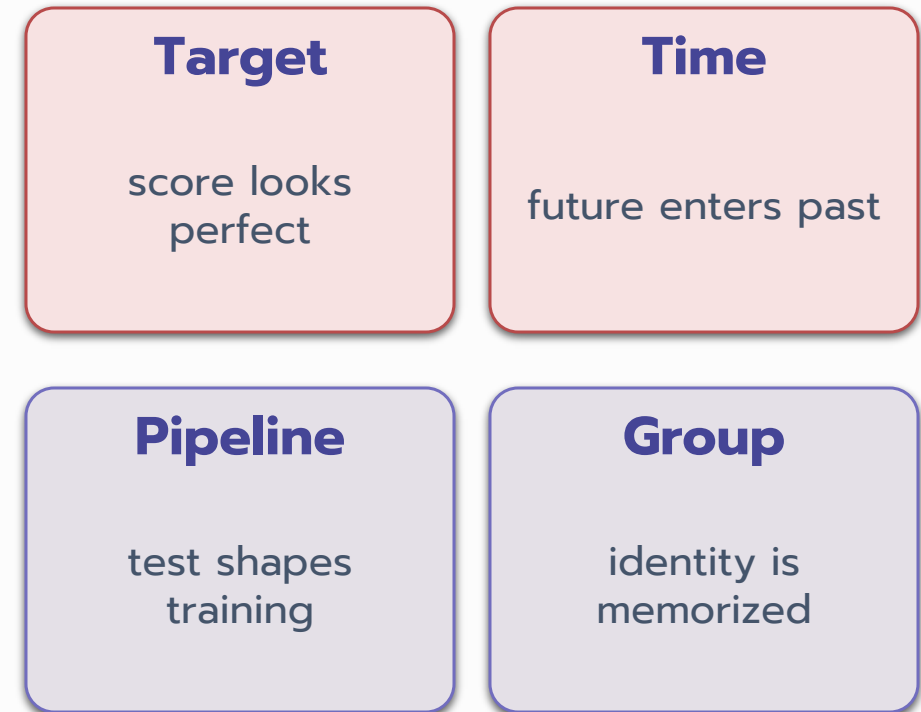
Traditional cross-tabs collapse under sparse combinations

- **Combinatorial growth:** 2 genders * 3 contracts * 4 payment types * 5 tenure bands = 120 cells.
- **Sparse cells:** many combinations contain zero or very few churners.
- **Multiple testing:** trying many tables inflates false discoveries unless corrected.
- **ML response:** regularize, validate out of sample, and record all tried features.



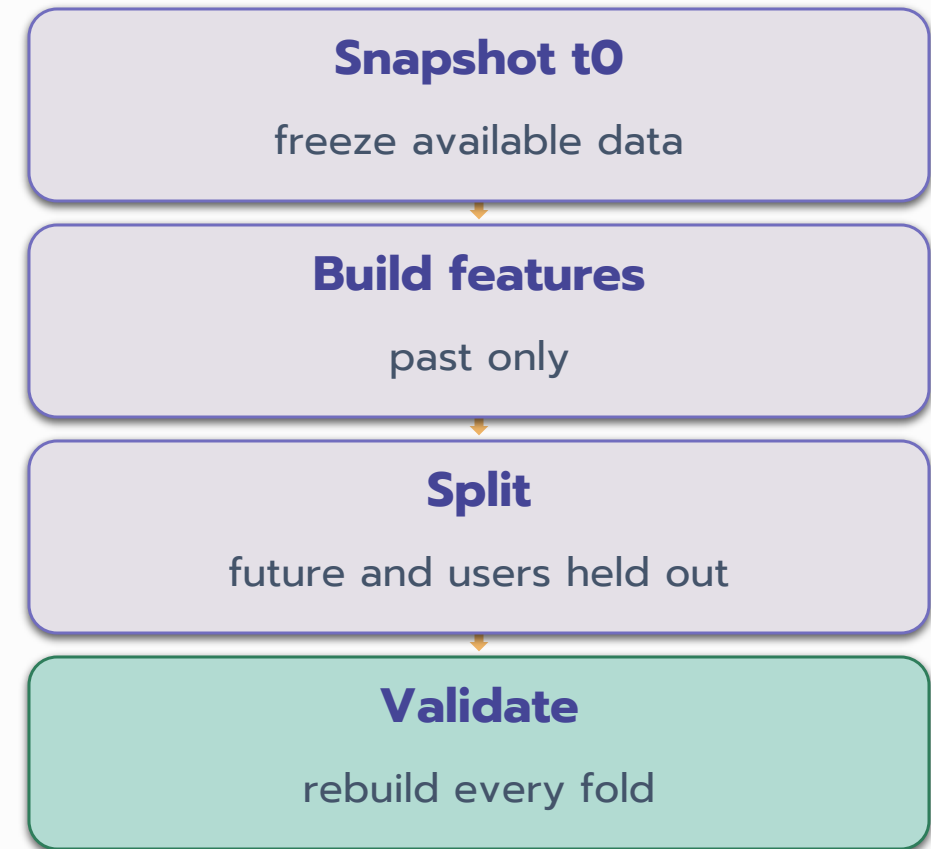
Leakage lets the model read information from the future

- **Target leakage:** a feature directly reveals churn, such as cancellation reason.
- **Temporal leakage:** events after t_0 appear in the feature window.
- **Preprocessing leakage:** scaling, encoding, or resampling uses test data.
- **Group leakage:** the same customer appears in both train and test.



Leakage detection starts with an as-of prediction test

- **Timestamp audit:** $\max(\text{feature_time}) < \text{prediction_time}$ for every row.
- **Split audit:** train dates precede validation dates; customer IDs do not cross groups.
- **Pipeline audit:** fit encoders, scalers, and resamplers only inside training folds.
- **Suspicion trigger:** an implausibly large validation jump demands investigation.



Module 3: Feature Engineering for Customer Behavior

MODULE 3

Summarize behavior without allowing future events to leak into the decision.

Choose windows

Aggregate events

Encode categories

Audit as-of time

RFM summarizes when, how often, and how much

- **Recency:** days since the most recent qualifying activity.
- **Frequency:** number of qualifying activities inside the window.
- **Monetary:** total or average value created inside the window.
- **Churn clue:** rising recency plus falling frequency often signals disengagement.

R

t0 - last activity

F

count(events)

M

sum(value)

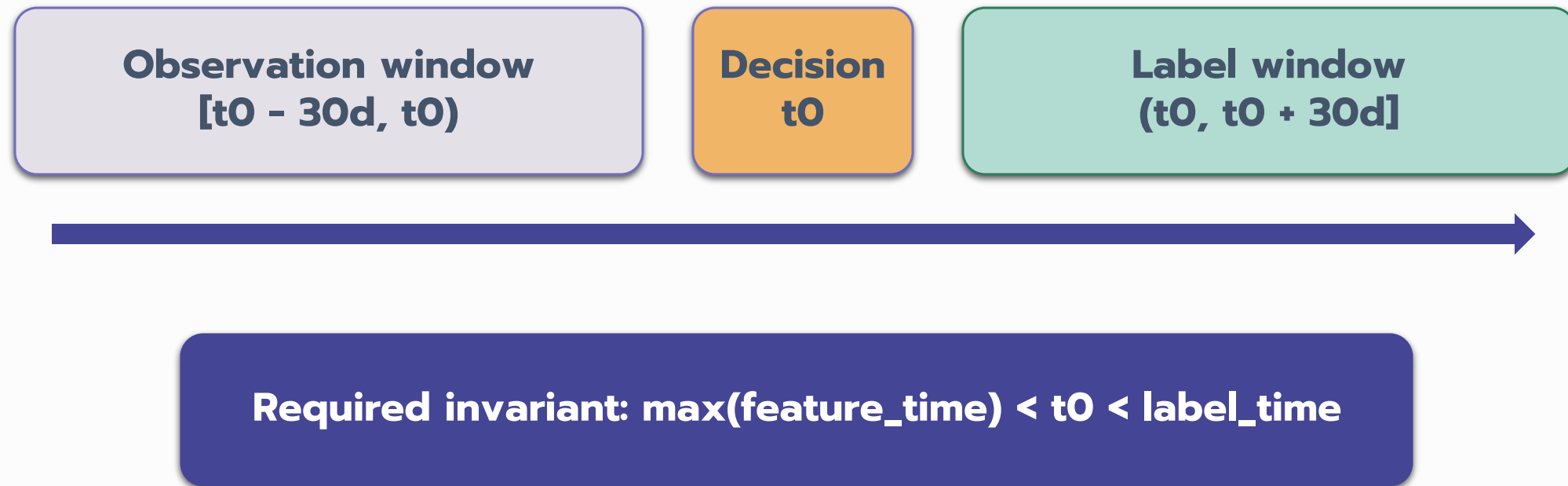
One customer becomes three auditable RFM features

- **C01:** last activity 14 days ago, 4 logins, \$85 value.
- **C02:** last activity 1 day ago, 22 logins, \$65 value.
- **C06:** last activity 21 days ago, 1 login, \$98 value.
- **Interpretation:** C06 is high-value but rapidly disengaging.

ID	R	F	M
C01	14	4	\$85
C02	1	22	\$65
C06	21	1	\$98

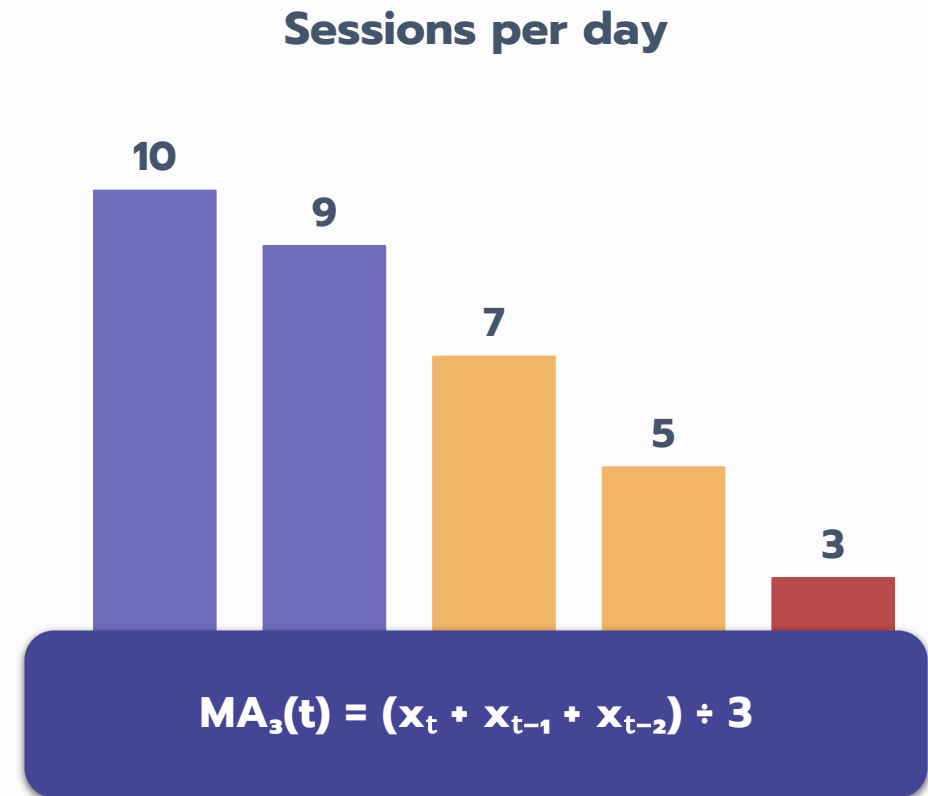
RFM(C06) = (21 days, 1 login, \$98)

A feature window must end before the decision moment



Rolling averages smooth noisy daily behavior

- **Daily sessions:** 10, 9, 7, 5, 3 show a steady decline.
- **Three-day mean:** the last value is $(7 + 5 + 3) \div 3 = 5$.
- **Model effect:** smoothing reduces one-day noise while preserving direction.



Trend features detect direction, not just level

- **Level feature:** mean sessions describes how much activity exists.
- **Trend feature:** a fitted slope describes the direction of change.
- **Business meaning:** negative engagement slope can trigger early service outreach.

$$\text{slope} = \frac{\sum(t - \bar{t})(x_t - \bar{x})}{\sum(t - \bar{t})^2}$$

+

activity rising

0

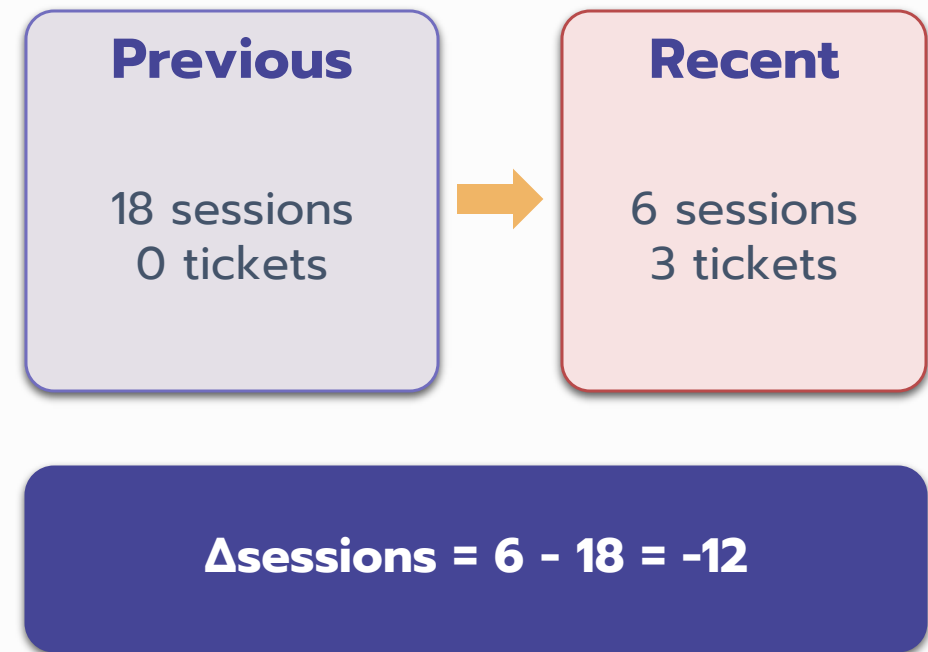
activity stable

-

activity falling

Behavioral shifts reveal a customer changing state

- **Previous 30 days:** 18 sessions and 0 support tickets.
- **Recent 30 days:** 6 sessions and 3 support tickets.
- **Shift:** sessions fell by 12, while support friction increased.
- **Model effect:** change features reveal state transitions earlier than totals.



One-hot encoding creates transparent category switches

- **Input:** Contract has Monthly, 1-year, and 2-year values.
- **Output:** each category receives its own 0/1 column.
- **Strength:** the meaning is visible and does not impose a fake order.
- **Risk:** new or rare categories require an explicit unknown policy.

Contract	Monthly	1-year	2-year
Monthly	1	0	0
1-year	0	1	0
2-year	0	0	1

Target encoding needs smoothing and out-of-fold computation

- **Raw estimate:** category churners ÷ category customers.
- **Smoothing:** small groups move toward the global churn rate.
- **Leakage control:** training rows receive out-of-fold encodings.
- **Test rule:** fit the mapping on all training data, then apply once.

$$TE(c) = (n_c \mu_c + \alpha \mu) \div (n_c + \alpha)$$

Fold A

fit on B + C



Fold B

fit on A + C



Fold C

fit on A + B

An as-of feature pipeline blocks future information



Invariant: $\text{feature_time} < t_0$ and customer groups never cross validation

Vectorized code builds leakage-safe features

```
df = events.sort_values(["customer_id", "event_time"])
cutoff = snapshots[["customer_id", "t0"]]
x = df.merge(cutoff, on="customer_id",
            validate="many_to_one")
x = x.loc[x["event_time"] < x["t0"]]
daily = x.groupby(["customer_id", "event_date"],
                 observed=True)
sessions = daily["sessions"].sum()
r7 = sessions.groupby(level=0).rolling(7,
min_periods=1).mean()
assert x["event_time"].lt(x["t0"]).all()
assert np.isfinite(r7.to_numpy()).all()
```

- 1 Sort once so time-based operations are deterministic.
- 2 Attach exactly one decision time to each customer snapshot.
- 3 Filter before aggregation; future rows never enter a window.
- 4 Vectorized grouping avoids iterative dataframe slicing; assertions fail closed.

Module 4: Foundational Predictive Modeling

MODULE 4

Estimate risk, explain the score, measure errors, and choose an action threshold.

Logistic score

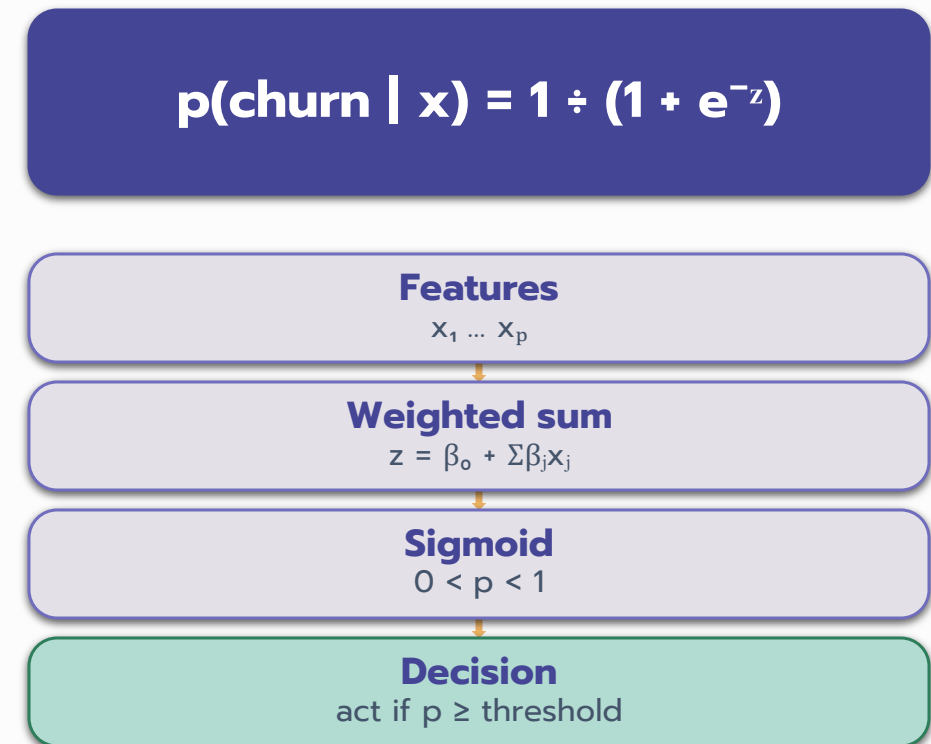
Tree rules

Error metrics

Business action

Logistic regression maps a weighted score to churn probability

- **Input x :** tenure, recency, contract, tickets, and other features.
- **Linear score z :** each feature contributes $\beta_j x_j$ to log-odds.
- **Sigmoid:** compresses any score into a valid probability.
- **Business use:** rank customers and apply a capacity-aware threshold.



Odds ratios translate coefficients into multiplicative risk

- **Formula:** odds ratio for feature j is $\exp(\beta_j)$.
- $\beta = \mathbf{0.69}$: odds multiply by $\exp(0.69) \approx 2.00$.
- $\beta = \mathbf{-0.51}$: odds multiply by $\exp(-0.51) \approx 0.60$.
- **Caution:** odds are not probabilities, and association is not causation.

$$\text{OR}_j = \exp(\beta_j)$$

$$\beta = 0.69$$

OR $\approx$ 2.00 · doubles odds

$$\beta = 0$$

OR = 1.00 · no odds change

$$\beta = -0.51$$

OR $\approx$ 0.60 · lowers odds

An illustrative customer receives a 61.8 percent churn score

- **Intercept:** -1.20 is the starting log-odds.
- **Monthly contract:** +0.80.
- **High recency:** +0.45; two or more tickets: +0.43.
- **Decision:** 61.8% exceeds a 60% teaching threshold, so review for action.

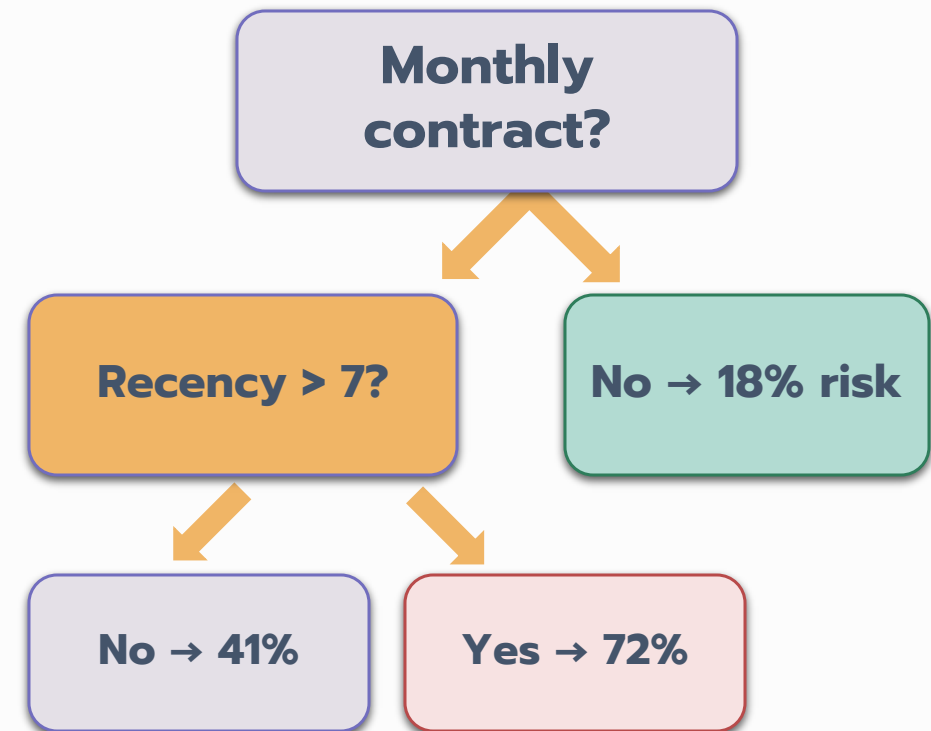
$$z = -1.20 + 0.80 + 0.45 + 0.43 = 0.48$$

$$p = 1 \div (1 + e^{-0.48}) = 0.618$$

61.8% risk → review high-value customer

A decision tree asks a sequence of yes-or-no questions

- **Split 1:** is the contract month-to-month?
- **Split 2:** is recency greater than 7 days?
- **Leaf:** estimate churn rate among training rows that reach it.
- **Control:** limit depth and minimum leaf size; validate on later customers.



The confusion matrix records four business outcomes

Actual / Predicted	Churn	Stay
Churn	TP = 18	FN = 6
Stay	FP = 12	TN = 64

TP
right offer

FP
wasted offer

FN
missed churn

TN
correct no action

Precision asks whether a retention offer is well targeted

- **Question:** of customers predicted to churn, how many truly churn?
- **Example:** 30 offers are sent; 18 reach true churners and 12 are false alarms.
- **Business meaning:** higher precision reduces wasted incentive cost.

$$\text{Precision} = \text{TP} \div (\text{TP} + \text{FP}) = 18 \div 30 = 60\%$$

30

predicted churners

18

true churners

60%

offer precision

Recall asks how many true churners we reached

- **Question:** of customers who truly churn, how many did we identify?
- **Example:** 24 customers churn; the model reaches 18 and misses 6.
- **Business meaning:** higher recall reduces missed retention opportunities.

$$\text{Recall} = \text{TP} \div (\text{TP} + \text{FN}) = 18 \div 24 = 75\%$$

24
true churners

18
reached

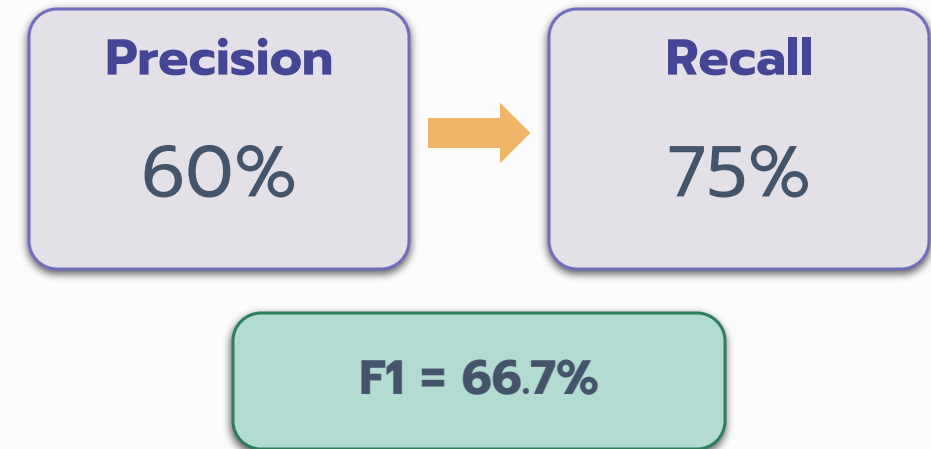
6
missed

75%
churn coverage

F1 balances precision and recall with a harmonic mean

- **Precision:** 60% of offers reach true churners.
- **Recall:** 75% of true churners are reached.
- **F1:** 66.7% summarizes both without favoring the larger value.
- **Limitation:** use expected profit when errors have different monetary costs.

$$F1 = 2PR \div (P + R) = 2(0.60)(0.75) \div 1.35 = 66.7\%$$



ROC-AUC measures ranking across every threshold

- **ROC curve:** plots true-positive rate against false-positive rate across thresholds.
- **AUC meaning:** 0.5 is random ranking; 1.0 is perfect ranking.
- **Business use:** compare ranking quality before selecting a contact threshold.
- **Limitation:** strong AUC can coexist with poor probability calibration.

$$\text{AUC} = P(\text{score}(\text{churner}) > \text{score}(\text{stayer}))$$

0.82 > 0.31
correct pair

0.67 > 0.22
correct pair

0.55 < 0.61
wrong pair

Choose actions from risk, value, and intervention cost

	LOW CLV	HIGH CLV
HIGH RISK	High risk · low CLV low-cost message or service fix	High risk · high CLV priority agent and tailored offer
LOW RISK	Low risk · low CLV no costly action; monitor	Low risk · high CLV protect experience; avoid over-contact

Customer lifetime value

$$\text{Expected net value} = p(\text{churn}) * p(\text{save} \mid \text{action}) * \text{CLV} - \text{action cost}$$

Module 5: Advanced ML and Imbalanced Data

MODULE 5

Increase model capacity while preserving honest validation and business alignment.

Ensemble trees

Handle imbalance

Validate by time

Choose by cost

Random forests average many decorrelated trees

- **Bootstrap rows:** each tree trains on a resampled customer dataset.
- **Random features:** each split considers only a subset of predictors.
- **Average:** many unstable trees combine into a more stable probability.
- **Risk:** deep correlated trees can still overfit and hide subgroup errors.

$$p_{\text{RF}}(\mathbf{x}) = (1 \div B) \sum_{\beta} p_{\beta}(\mathbf{x})$$

Tree 1

0.72

Tree 2

0.58

Tree 3

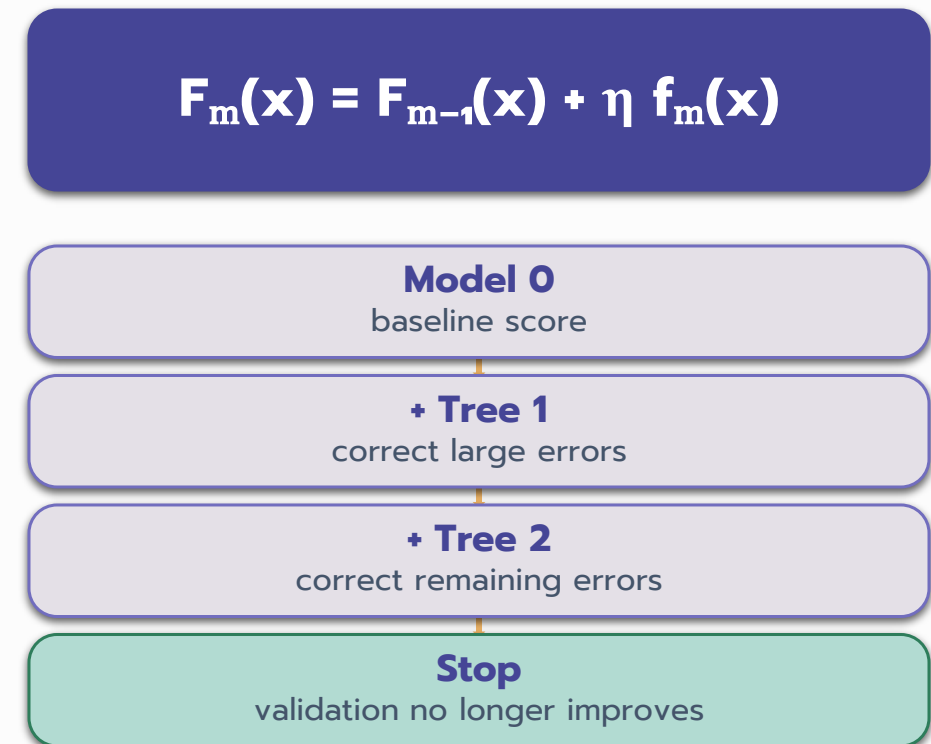
0.66

Average

0.65 risk

Gradient boosting corrects one model error at a time

- **Start:** make a simple probability prediction for every customer.
- **Measure error:** identify where the current model is too high or too low.
- **Add a tree:** fit a small correction to those errors.
- **Control:** learning rate, depth, and early stopping limit overfitting.



Compare XGBoost and LightGBM fairly

Dimension	XGBoost	LightGBM	Business implication
Tree growth	level-wise by default	leaf-wise	LightGBM may fit complex structure quickly
Large data	strong general baseline	histogram-based efficiency	benchmark time and memory
Overfitting control	depth, regularization	leaf count, min leaf data	early stop on later data
Selection	same folds and metric	same folds and metric	choose reproducible net value

Fair comparison = identical data * identical folds * identical metric

Imbalance makes accuracy look better than the model

- **Customers:** 950 stay and 50 churn.
- **Naive rule:** predict stay for every customer.
- **Result:** 950 correct predictions but TP = 0 and FN = 50.
- **Business failure:** no at-risk customer receives a retention action.

Accuracy

$$950 \div 1,000 = 95\%$$

Churn recall

$$0 \div 50 = 0\%$$

Class weights make costly churn errors count more

- **Mechanism:** multiply each training loss by its class weight.
- **Balanced rule:** rarer classes receive larger weights.
- **Strength:** keeps original rows and works inside most classifiers.
- **Control:** select weights within validation, not from the test result.

$$w_c = N \div (K * N_c)$$

Stay weight

$$1,000 \div (2 * 950) = 0.526$$

Churn weight

$$1,000 \div (2 * 50) = 10.0$$

SMOTE interpolates new minority examples between neighbors

- **Choose:** a churn customer x_i and a nearby churn customer x_{nn} .
- **Interpolate:** sample λ between 0 and 1 to create a point on the segment.
- **Leakage control:** resample after the training split, separately inside each fold.
- **Risk:** unsafe distance metrics can create unrealistic mixed-category rows.

$$x_{\text{new}} = x_i + \lambda(x_{nn} - x_i), 0 \leq \lambda \leq 1$$



Synthetic point stays between observed minority neighbors

ADASYN generates more examples near difficult boundaries

Difficulty r_i : fraction of majority examples among K neighbors.

Allocation: harder minority points receive more synthetic samples.

Strength: focuses model attention near an uncertain boundary.

Risk: outliers and mislabeled rows may receive too much attention.

$$r_i = \text{majority neighbors} \div K; g_i \propto r_i$$

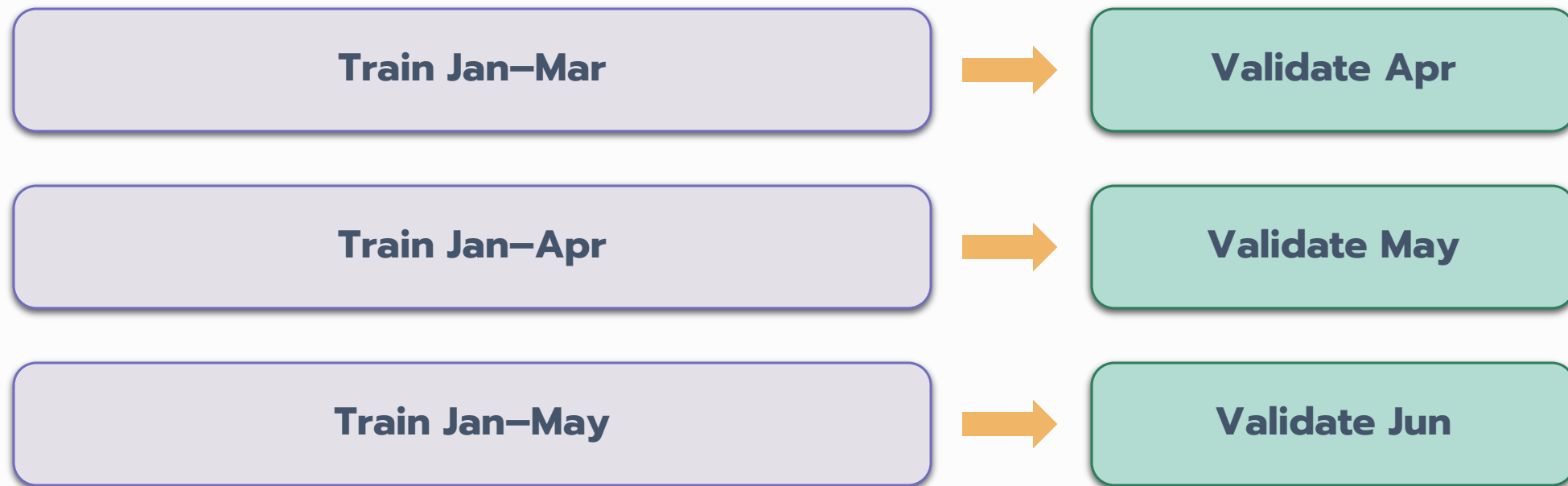
Easy region

1 of 5 majority
neighbors → few
new points

Hard region

4 of 5 majority
neighbors → more
new points

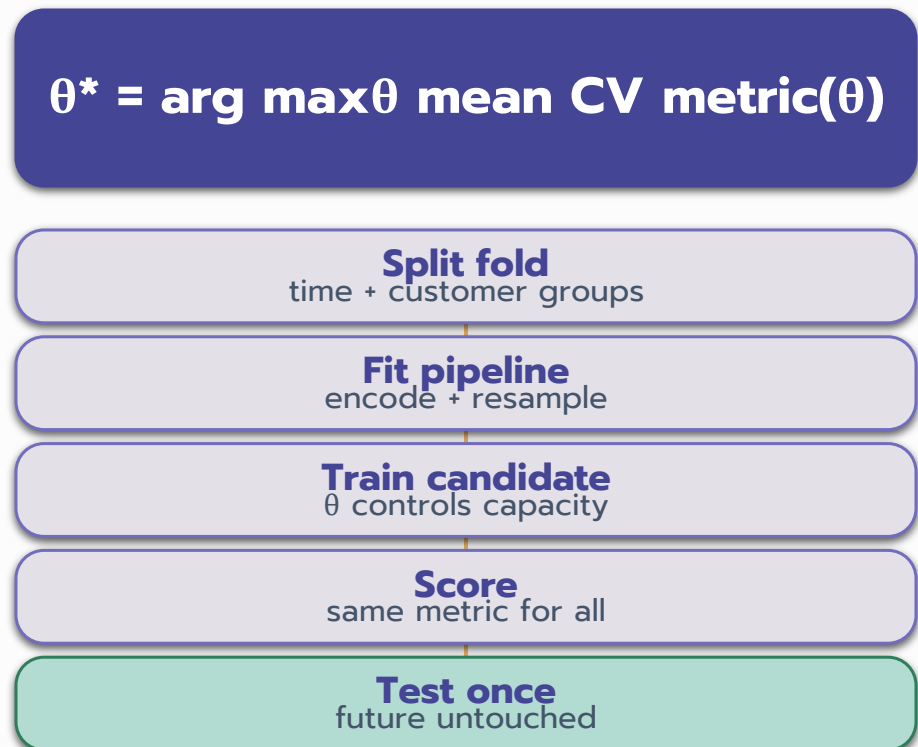
Cross-validation must respect both time and customer identity



Group invariant: customer IDs in train $\cap$ validation = $\emptyset$

Tune model capacity inside a leakage-safe pipeline

- **Candidate θ :** depth, learning rate, leaf size, and class weight.
- **Inner folds:** fit encoders and resampling separately for each candidate.
- **Selection:** choose the preregistered metric and record every trial.
- **Final gate:** evaluate once on the untouched future test period.



Match the method to data risk and business cost

Situation	Preferred starting method	Validation rule	Retention strategy
Small, explainable baseline	logistic + class weights	forward grouped folds	clear odds and thresholds
Nonlinear interactions	random forest	limit depth and leaf size	segment by stable drivers
Large tabular data	XGBoost / LightGBM	early stop on later folds	rank by expected net value
Very rare churn	weights, SMOTE, or ADASYN	resample inside fold only	optimize recall under capacity

Stop rule: reject any model whose future-period value or subgroup safety is worse than the baseline.

Module 6: Actionable Intelligence and Causal Inference

MODULE 6

Move from who may leave to which intervention creates incremental retention.

Estimate risk

Estimate uplift

Randomize action

Monitor effect

Prediction finds risk; causation finds what changes outcomes

- **Risk:** $P(\text{churn} \mid X)$ identifies customers likely to leave.
- **Uplift:** $E[Y(1) - Y(0) \mid X]$ estimates the incremental treatment effect.
- **Why different:** the highest-risk customer may be impossible to persuade.
- **Identification:** random assignment or credible causal assumptions are required.

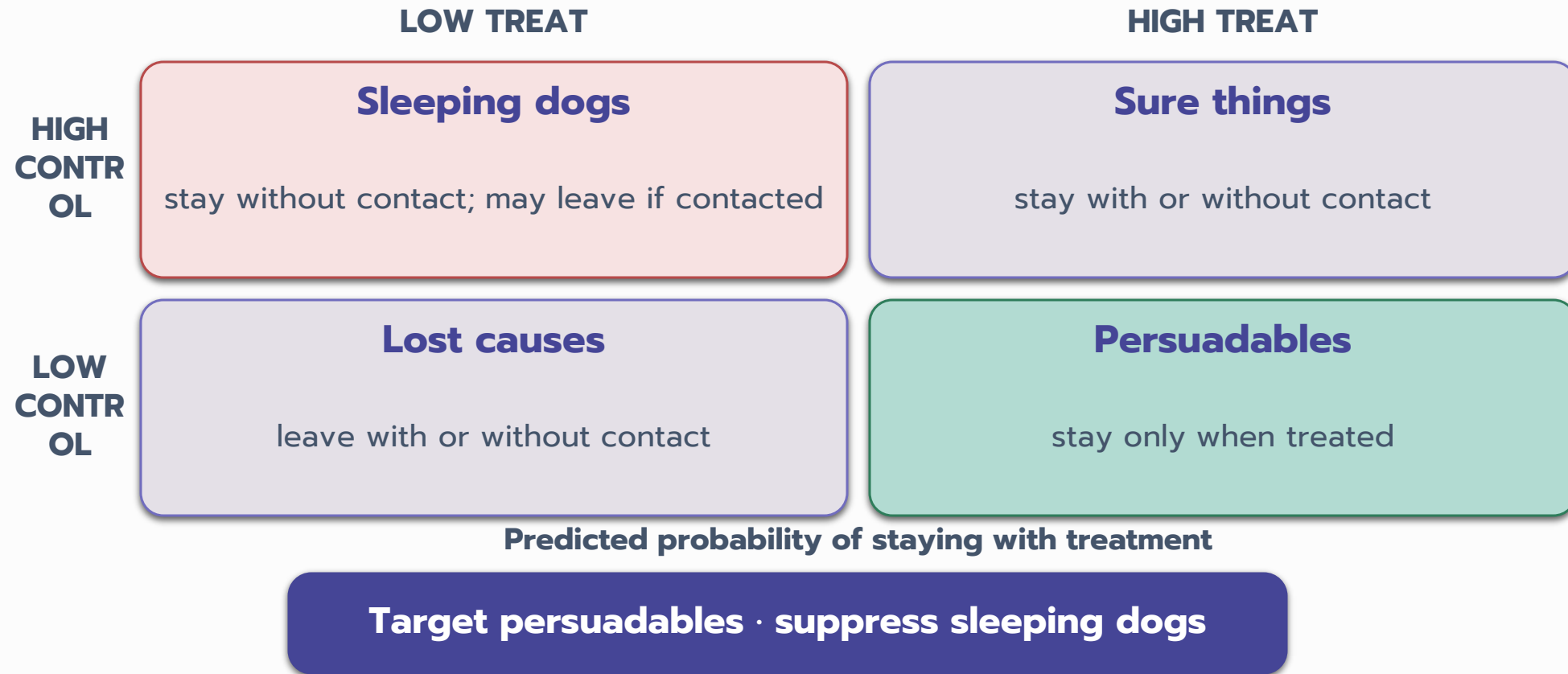
Predictive question

Who is likely to churn?
 $\text{Risk} = P(Y = 1 \mid X)$

Causal question

Who stays because we act?
 $\tau(X) = E[Y(1) - Y(0) \mid X]$

Uplift modeling separates four response types



Two models estimate the incremental effect of contact

- **Treatment model:** estimate $P(\text{stay} \mid X, T = 1)$.
- **Control model:** estimate $P(\text{stay} \mid X, T = 0)$.
- **Uplift score:** subtract control probability from treatment probability.
- **Integrity:** use randomized or causally identified treatment data; validate uplift out of sample.

Treatment model

$$p_1(x) = 0.84$$

Control model

$$p_0(x) = 0.70$$

$$\text{uplift}(x) = p_1(x) - p_0(x) = +0.14$$

+14 percentage points → promising persuadable

A randomized A/B test measures retention lift

- **Randomize:** 5,000 receive the retention action and 5,000 receive business as usual.
- **Observe:** 82% retention in treatment versus 78% in control.
- **Pre-register:** primary metric, horizon, sample size, exclusions, and stop rule.
- **Inference:** use customer-clustered or HAC errors when outcomes repeat over time.

$$\text{Lift} = 0.82 - 0.78 = +0.04 = +4.0 \text{ pp}$$

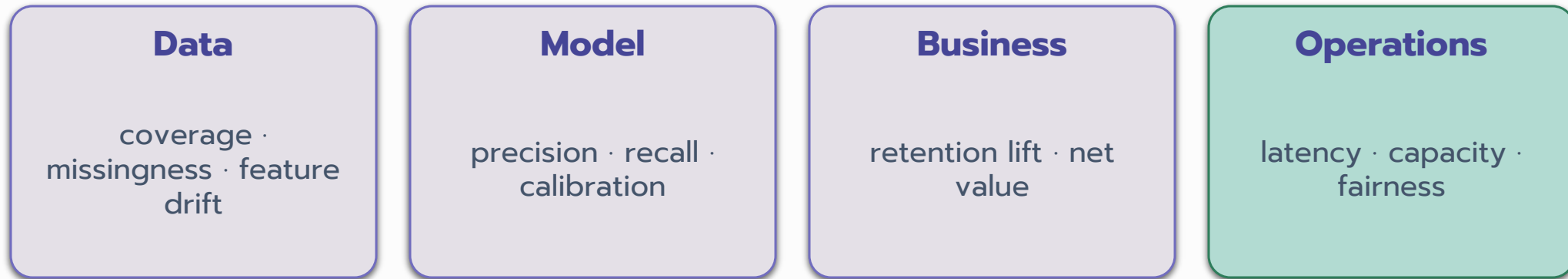
Standard error

$$\sqrt{.82(.18)/5000 + .78(.22)/5000} \approx 0.8 \text{ pp}$$

95% interval

$$+4.0 \pm 1.96(0.8) = [2.4, 5.6] \text{ pp}$$

Monitor data, model, business, and treatment effects



Concept drift: $P(Y | X)$ changes even when the scoring code does not

Stop-loss: pause treatment when measured uplift turns negative, safety fails, or drift exceeds the approved threshold.

Tip and Trick: predict, value, test, and monitor

1. Predict

Who may churn?

2. Value

Who is worth saving?

3. Test

What action causes
lift?

4. Monitor

Does value persist?

Best rule: act only when expected incremental value is positive and validated

Customer Churn

961731 Business Data Analytics

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